JAIWAL PATEL

Aspiring Computer Science Engineer

PORTFOLIO LINK Dubai, UAE

SUMMARY

Grade 12 student with a strong foundation in mathematics and computer science, combining independent advanced study with published AI/ML research, software development, teaching, and community leadership. Founded EcoRevive Dubai and built EcoRevive OS, a cloud-native platform that digitized its manual operations. Seeking to pursue computer science at the undergraduate level, with an emphasis on building technology that creates meaningful impact.

EDUCATION

Grade 12 (CBSE) May ’27

GEMS New Millennium School, Dubai, UAE

Grade 11: 95.8%; Grade 10: 96.16%; Grade 9: 88.5%

SAT: 1580 on first attempt (Math 800; Reading & Writing 780)

SKILLS

Programming Languages & Technologies: Python, JavaScript, TypeScript, HTML, CSS, Django/DRF, React, Bootstrap, PostgreSQL, Redis, Docker, Git/GitHub, JSON, pytest, Azure, Amazon Lex, CNN, Deep Learning, Testing & Debugging

Courses & Certifications: AP Calculus BC (5); AP Statistics (5); AP Physics C: Mechanics (5); AP Physics C:Electricity & Magnetism (5); AP Computer Science A (4); Harvard CS50P; Cisco Networking Basics; Statistical Learning with R

Analytical Skills: Mathematical Problem-Solving, Statistical Analysis, Introductory Machine Learning

Note: School offers no AP courses; all five APs were pursued independently.

ACADEMIC AND RESEARCH PROJECTS

Retinal OCT Image Classification *Jun '25 - Present*

Student Researcher & Co-author, [JEEEMI journal], https://jeeemi.org/index.php/jeeemi/article/view/1991

• Co-authored and published a study using a lightweight CNN-SE model for retinal OCT classification, reporting 99% accuracy on OCT2017 and 97% on OCT-C8.

• Contributed to the mathematical foundation, abstract analysis, manuscript drafting and editing, project execution, and controlled ablation studies under Professor Parth Dave, who led model implementation.

EcoRevive OS *Apr '26 - Aug '26*

Product Developer, https://ecorevive.app

• Transformed manual e-waste operations into a scalable cloud-native platform, digitizing request-to-handover workflows with AI-assisted tools.

• Defined requirements and product and architecture decisions; reviewed AI-assisted code, tested workflows, and debugged failures.

• Built with Django/DRF, PostgreSQL, Redis, React, TypeScript, Vite, Docker Compose, pytest, and an Azure VM.

Student Book Exchange *Jan '26 - Mar '26*

Founder & Developer

• Built a Django book-exchange platform with ~100 listings, enabling the exchange of 47 books among 22 launch users.

• Trained 2 coordinators to continue running the platform.

• School wide adoption planned for 2027.

AI Hotel Concierge Bot *Jul '25*

Amazon Web Services (AWS) AI/ML Program

• Built a 24/7 concierge bot in Amazon Lex handling room service, spa and table bookings, housekeeping/laundry, and local recommendations.

EXPERIENCE

EcoRevive Dubai *Dec '24 - Present*

Founder

• Coordinated 450+ pickups from ~350 families and diverted ~3,500 kg of e-waste through certified recycler Enviroserve Dubai.

• Mobilized volunteers and identified a successor to help carry the initiative forward.

NMS GATE Club / Aqua Shield *Apr '25 - Mar '26*

President & Innovation Team Lead

• Led a ~23-member GATE Club and organized 3 major events.

• Led a 5-student Aqua Shield team and presented at the Innovation Fair among ~20-25 teams.

Academic & Coding Mentorship *Apr '25 -Dec '25*

Math, SAT & Coding Mentor Grade 11

• Mentored 5 junior students in math; led a coding bootcamp for 8 students and an SAT bootcamp for 15 (6 underprivileged), improving SAT scores by ~100-150 points.

Amazon Web Services (AWS) AI/ML Program *Jul 6 - Jul 27, '25*

AI Intern

• Completed a 24-hour hands-on program with an AWS mentor, building an AI concierge bot in Amazon Lex.

Neokul Virtual Global Apprenticeship *Mar '25 - May '25*

AI Apprentice

• Completed a 30-hour AI apprenticeship and built a Python/JSON homework chatbot for students.

Mashauri Virtual Global Apprenticeship *Jul '25 - Sep '25*

Software Development & Engineering Apprentice

• Completed a 130-hour apprenticeship and built a responsive web application using JavaScript and Bootstrap.

HONORS & AWARDS

• AP Scholar with Distinction Award *2026*

•International Olympiad Foundation Mathematics Olympiad, Round 2 Qualifier, UAE Rank 4 *2025*

• TALLENTEX Overseas 2025 (ALLEN), Country Rank 5 (UAE), Overseas Rank 11 *2025*

• CBSE Aryabhata Ganit Challenge, Regional Top 100 Grade 10 *2024*

• Ei ASSET Maths, National Level Topper, UAE Grade 9 *2024*

• SOF National Science Olympiad, Gold Medal of Distinction; Zonal Rank 17 *2024*

INTEREST & EXTRACURRICULARS

Competitive Chess *Jan '23 - Present*

• Trained with a chess coach and competed in 6 FIDE-rated tournaments, reaching a rating of 1409; play online to sharpen calculation and decision-making.

Interests: Mathematics, reading, chess, running, software building, statistics, sustainability, and volunteering.